Improving Spatial Generalization with Edge-Gated Spatiotemporal Graph Neural Networks for Temperature Inversion Forecasting (Supplementary Material)

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A Appendix

A.1 Hyperparameters

We include the hyperparameters for each baseline model in Tables 1, 2, 3, and 4. All models were tuned via grid search using the validation set RMSE as the signal in the cluster-based CV setting. These measures the more difficult generalization setting for entire regions and is much less computationally expensive than leave-one-station-out. These tuned values are then used for the leave-one-station-out runs. GraphWaveNet deviates slightly from the default settings in the Torch Spatiotemporal package (Cini and Marisca 2022) we used for implementation, and did not appear very sensitive to the tuning.

Table 1: Hyperparameters of the LSTM model.

Parameter	Value
LSTM hidden units	64
LSTM layers	2
CNN embedding dimension	16
Dropout	0.15
Learning rate	3×10^{-4}
Batch size	32
Max epochs	25
Teacher forcing (TF)	LinearAnneal(0.9, 0.1)
TF Schedule	6 epochs

Table 2: Hyperparameters of the DKL-LSTM model.

Parameter	Value
LSTM hidden units	64
LSTM layers	2
CNN embedding dimension	16
Dropout	0.3
GP kernel	RBF
Inducing points	128
Learning rate	3×10^{-4}
GP learning rate	10^{-2}
Batch size	32
Max epochs	25

Table 3: Hyperparameters of the GraphWaveNet model.

Parameter	Value
Temporal layers	12
Hidden channels	32
Feed-forward dimension	256
Node embedding size	10
Readout latent dimension	32
Dropout	0.3
Learning rate	10^{-3}
Batch size	32
Max epochs	25

Table 4: Hyperparameters of the EGP-GNN model.

Parameter	Value
GRU hidden units	64
GRU layers	2
Edge feature dimension	32
Attention hidden dimension	32
CNN embedding dimension	16
Dropout	0.2
Learning rate	10^{-3}
Batch size	32
Max epochs	25
Teacher forcing (TF)	LinearAnneal(0.9, 0.1)
TF Schedule	6 epochs

A.2 Leave-one-station-out Stations Won

Table 5 shows the number of folds (i.e. stations) for which an algorithm outperforms all other algorithms (denoted as the "stations won") across the leave-one-station-out cross-validation experiments. We include both EGP-GNN and EGP-GNN (RF seed). For the EGP-GNN (RF seed) column, we show how many folds are won out of the three baselines, replacing EGP-GNN for comparison. The majority of stations, across all considered horizons, are won by EGP-GNN even when the RF seed is used. When EGP-GNN is not the best performer for a fold, there is not a consensus second-best performer among the baselines.

We present in the following sections some commentary

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and figures (see Figures 1 and 3) that cover some of the heterogeneity present across local station installations. Many of the stations and regions are more challenging for all the models, whether based on specific characteristics or simply poorer quality historical weather variable measurements. Additionally, different modeling assumptions we have seen vary in performance on different regions and stations, a full presentation of which we leave for future work.

A.3 Cluster-Based CV Per Fold

We include a breakdown of the cluster-based cross-validation on a per-fold basis in Table 6. This highlights the consistency of EGP-GNN across the different folds, over the 3 time horizons and the total aggregate. We show the worst-case fold behavior and the range as well, on which EGP-GNN achieves the lowest RMSE. The EGP-GNN (RF seed) is included in the table to show the performance of this against the baselines. Lastly is shown a count of folds won by each model outright among the three baselines and EGP-GNN - for EGP-GNN (RF seed), this represents folds won against the three baselines. We can see that EGP-GNN wins the majority of folds across all horizons, and EGP-GNN (RF seed) does not deviate from this.

More challenging clusters for EGP-GNN to generalize across are those that generally have cases where nearest spatial neighbors within the cluster are less representative and predictive of the inversion behavior at a target station. For example, Fold 0 consists of stations split across different sides of Puget Sound. A full map of the stations by cluster is shown in Figure 1.

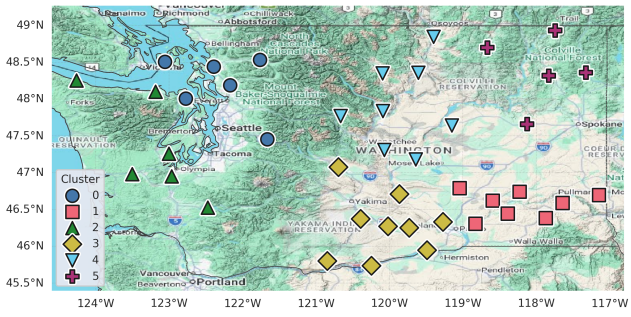


Figure 1: Mesonet tower weather station locations across Washington State, partitioned and labeled by cluster used for the cluster CV analysis.

A.4 GraphWaveNet Inversion Strength Feature Comparison

Here, we compare briefly against GraphWaveNet once again, which is generally the second best performing model across all experiments. We allow GraphWaveNet to consume the true previous inversion strength value $y_i(T-1)$ at each timestep during forecasting as a future covariate. This replicates a "perfect seeding" for GraphWaveNet. Figure 2 shows the averaged RMSE value at each time horizon for both models, as well as a listing of number of "stations/folds won" between the two models, where stations/folds won again indicates a lower value on the averaged RMSE over time horizons within a hold-out station or fold respectively.

EGP-GNN achieves a higher number of stations won on the leave-one-station-out (LOO) CV setting and has a lower aggregate RMSE when averaged over all hold-out sets and horizons (equivalent to the All Horizons RMSE elsewhere). This holds true for the cluster-based CV as well when considering the 6 folds. The anchoring term in EGP-GNN provides a substantial benefit on shorter time horizons (out to approximately 24 h) in both CV regimes. On the leave-one-station-out setting, this benefit continues to hold marginally out to 48 h. In the cluster CV case, the models are nearly indistinguishable from 25 h until 43 h. The exact 1 day and 2 day ahead predictions at 24 h and 48 h do seem to benefit slightly from repeated access to the true $y_i(T-1)$, likely caused by time-of-day similarity.

We note that when "seeding" the values for both EGP-GNN and GraphWaveNet using the considered Random Forest model, the differences are more pronounced in favor of EGP-GNN suggesting it is less reliant on perfect seeding.

A.5 Station Static Feature Distributions

To illustrate the heterogeneity across the tower stations in our analysis, we include a boxplot of several common static features that define the topography of the stations, along with fractional counts of the NLCD land cover features for the stations' (4km x 4km) regions on a per-station basis. These are shown in Figure 3.

These figures show the stations have a wide spread of static terrain features along with (in many cases) substantially varying dominant land cover types. This makes generalization across regions a more difficult task than individual stations if the model has trained on the majority of the state. It may prove useful to benchmark different spatial generalization by defining hold-out clusters by similarities in terrain or land cover profiles, but we also leave this for future work.

A.6 Station Information

For completeness, Table 7 shows the stations used for this analysis, along with elevation and available data for each. After preprocessing and imputation, all stations used have consistent data measurements at 1 hr intervals between the first and last reported measurement dates here.

Table 5: Leave-one-station-out (LOOCV) test RMSE, mean ± 95% CI across the 42 held-out stations. Lower is better; the best model in each row is bold. The "stations won" row counts, per horizon, the held-out stations in which each model has the lowest RMSE (outright wins among the observed-seed models); the RF-seed column instead counts stations in which it beats every baseline (LSTM, DKL-LSTM, GraphWaveNet).

Forecast Horizon	LSTM	DKL-LSTM	GraphWaveNet	GNN	GNN (RF seed)
1 h	1.412 ± 0.130	1.407 ± 0.125	1.410 ± 0.121	$\mathbf{0.886 \pm 0.079}$	1.240 ± 0.108
stations won	$0/42$	$0/42$	$0/42$	$\mathbf{42/42}$	$33/42$
24 h	1.533 ± 0.128	1.500 ± 0.128	1.509 ± 0.127	$\mathbf{1.417 \pm 0.116}$	1.456 ± 0.124
stations won	$2/42$	$8/42$	$10/42$	$\mathbf{22/42}$	$26/42$
48 h	1.570 ± 0.129	1.518 ± 0.129	1.529 ± 0.130	$\mathbf{1.481 \pm 0.119}$	1.499 ± 0.123
stations won	$4/42$	$10/42$	$9/42$	$\mathbf{19/42}$	$19/42$
All Horizons	1.527 ± 0.129	1.495 ± 0.129	1.506 ± 0.126	$\mathbf{1.418 \pm 0.117}$	1.450 ± 0.124
stations won	$4/42$	$4/42$	$7/42$	$\mathbf{27/42}$	$26/42$

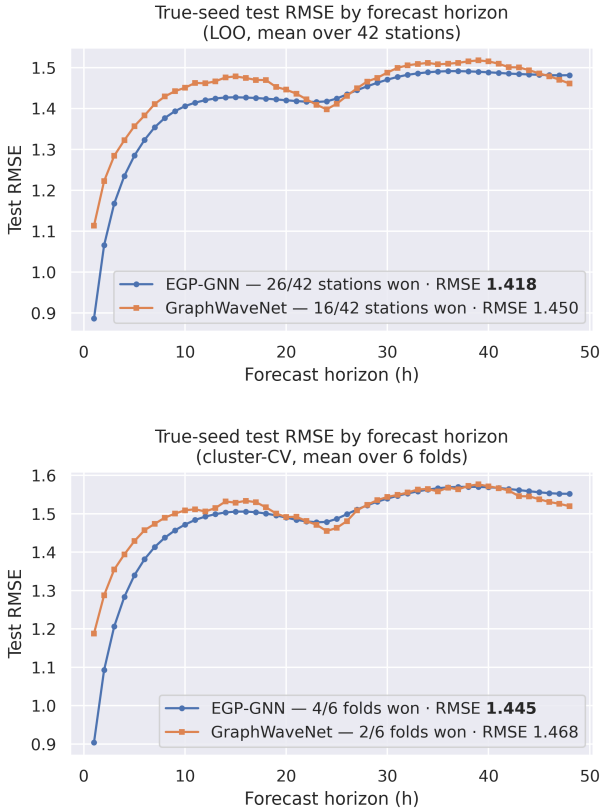


Figure 2: A per-horizon breakdown of the test RMSE with EGP-GNN and GraphWaveNet when GraphWaveNet is given the true inversion strength $y_i(T-1)$ as a future covariate for each time horizon. **Top:** The leave-one-station-out (LOO) cross-validation setting. **Bottom:** The cluster-based cross-validation setting. Stations/folds won denotes how many stations or folds out of the total each model achieves the lowest averaged RMSE over, and the printed RMSE is equivalent to the All Horizons RMSE, averaged over 1 h – 48 h.

Table 6: Per-fold leave-one-cluster-out test RMSE. Each row is one held-out spatial cluster (n = held-out stations); the 1, 24, and 48 h blocks are pooled over that fold's held-out stations at that lead time, and the All Horizons block averages that fold's held-out stations equally across all 48 horizons. Lower is better; the best model in each row is bold. The 'folds won' row counts, per lead time, the folds in which each model has the lowest RMSE (outright wins among the models shown); the RF-seed column instead counts folds in which it beats every baseline (LSTM, DKL-LSTM, GraphWaveNet).

Horizon	Fold	n	LSTM	DKL-LSTM	GraphWaveNet	EGP-GNN	EGP-GNN (RF seed)
1 h	0	6	1.279	1.258	1.208	**0.719**	1.133
	1	8	1.442	1.400	1.441	**1.001**	1.289
	2	6	1.164	1.193	1.465	**0.618**	1.113
	3	9	2.038	2.652	1.581	**1.099**	1.481
	4	8	1.460	1.771	1.410	**0.934**	1.251
	5	5	1.885	1.906	1.699	**1.052**	1.479
	mean ± 95% CI		1.545 ± 0.361	1.697 ± 0.574	1.467 ± 0.174	$\mathbf{0.904 \pm 0.203}$	1.291 ± 0.169
	worst fold		2.038	2.652	1.699	**1.099**	1.481
	range		0.874	1.459	0.491	0.482	**0.368**
	folds won		0/6	0/6	0/6	**6/6**	6/6
24 h	0	6	1.280	**1.277**	1.301	1.296	1.391
	1	8	1.562	**1.555**	1.557	1.578	1.592
	2	6	1.272	1.255	1.348	**1.218**	1.245
	3	9	2.290	2.706	1.774	**1.683**	1.706
	4	8	1.587	1.800	1.500	**1.431**	1.500
	5	5	2.159	1.935	1.859	**1.666**	1.773
	mean ± 95% CI		1.692 ± 0.457	1.755 ± 0.567	1.557 ± 0.235	$\mathbf{1.479 \pm 0.205}$	1.535 ± 0.207
	worst fold		2.290	2.706	1.859	**1.683**	1.773
	range		1.018	1.451	0.558	**0.465**	0.528
	folds won		0/6	2/6	0/6	**4/6**	4/6
48 h	0	6	1.290	**1.281**	1.314	1.410	1.502
	1	8	**1.591**	1.611	1.632	1.651	1.640
	2	6	1.311	1.273	1.356	1.276	**1.270**
	3	9	2.325	2.668	1.820	**1.718**	1.730
	4	8	1.592	1.814	1.506	**1.483**	1.530
	5	5	2.268	1.949	1.860	**1.775**	1.850
	mean ± 95% CI		1.729 ± 0.481	1.766 ± 0.546	1.581 ± 0.241	$\mathbf{1.552 \pm 0.203}$	1.587 ± 0.212
	worst fold		2.325	2.668	1.860	**1.775**	1.850
	range		1.035	1.395	0.545	**0.499**	0.579
	folds won		1/6	2/6	0/6	**3/6**	3/6
All Horizons	0	6	1.243	**1.242**	1.248	1.286	1.385
	1	8	**1.506**	**1.506**	1.526	**1.506**	1.508
	2	6	1.240	1.226	1.365	**1.173**	1.223
	3	9	2.046	2.226	1.688	**1.649**	1.665
	4	8	1.560	1.784	1.497	**1.438**	1.493
	5	5	2.070	1.844	1.788	**1.616**	1.636
	mean ± 95% CI		1.611 ± 0.389	1.638 ± 0.407	1.518 ± 0.209	$\mathbf{1.445 \pm 0.196}$	1.485 ± 0.172
	worst fold		2.070	2.226	1.788	**1.649**	1.665
	range		0.830	1.000	0.540	0.476	**0.441**
	folds won		0/6	2/6	0/6	**4/6**	4/6

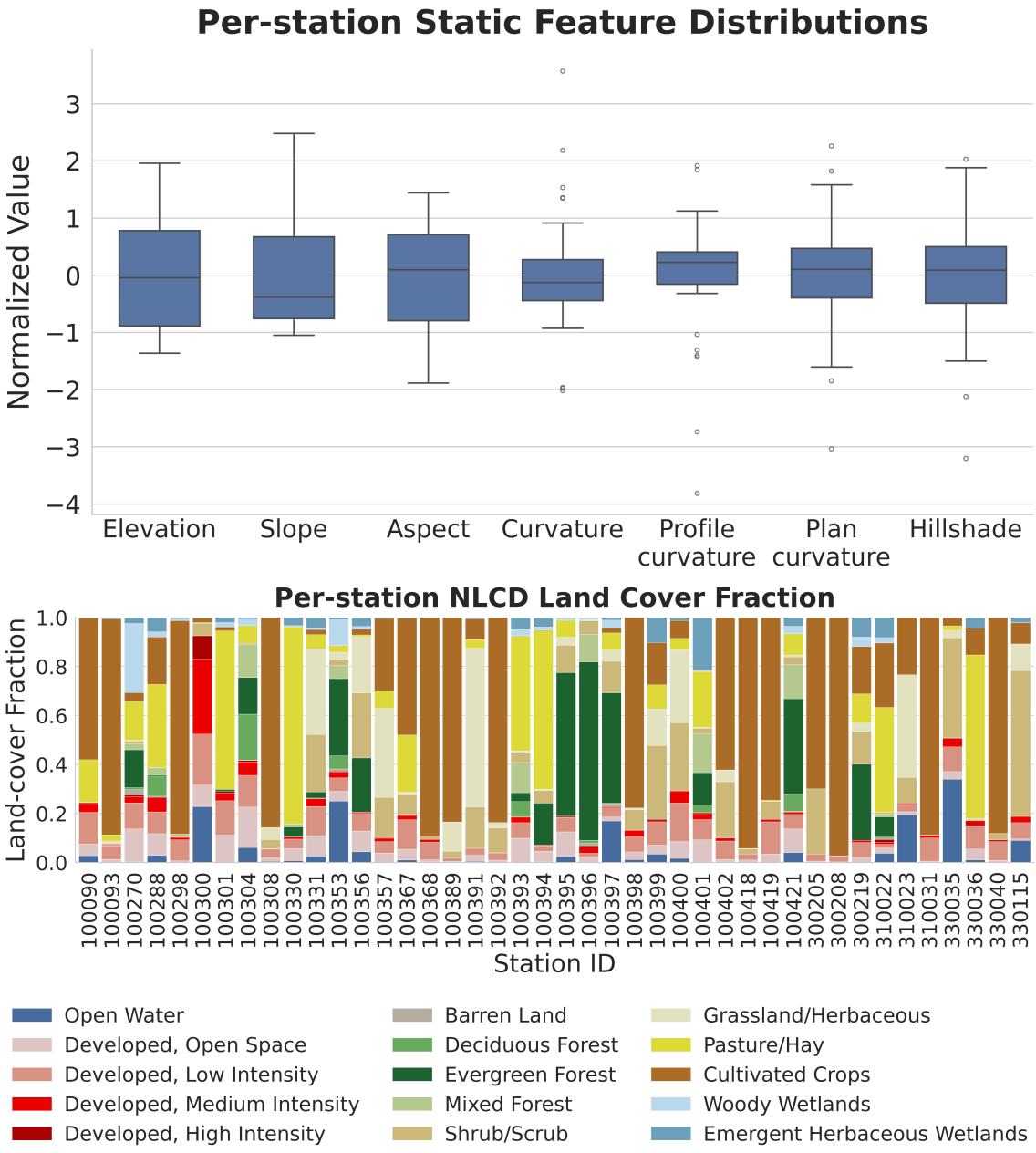


Figure 3: *Top:* Distributions of selected static terrain features (normalized) across the stations presented as box plots. The lines show the 25% and 75% quartiles. *Bottom:* Fractions of the land cover types at each station. Note the values for both the static terrain features and land cover vary heavily across different stations and regions, making generalization a challenge.

Table 7: The 42 AgWeatherNet tower weather stations used in the experiments, with elevation and date ranges for the available data.

ID	Station Name	Elev. (ft)	First Measurement	Last Measurement
100090	Mount Vernon	16	10/26/2023	08/31/2025
100093	Pullman	2,545	05/24/2023	08/31/2025
100270	Tumwater	186	06/27/2024	08/31/2025
100288	Arlington	46	10/26/2023	08/31/2025
100298	Othello	1,200	11/02/2021	08/31/2025
100300	Richland	376	10/13/2021	08/31/2025
100301	Sequim	138	07/17/2024	08/31/2025
100304	Concrete	254	09/14/2023	08/31/2025
100308	Big Flat	730	11/02/2023	08/31/2025
100330	Friday Harbor	38	10/26/2023	08/31/2025
100331	Twisp	1,569	10/13/2023	08/31/2025
100353	Clallam Bay	53	09/22/2022	08/31/2025
100356	Republic	2,357	09/21/2023	08/31/2025
100357	Goldendale	1,673	04/19/2023	08/31/2025
100367	Prosser	830	06/05/2024	08/31/2025
100368	Mattawa	784	04/27/2023	08/31/2025
100389	Pomeroy	1,742	05/24/2023	08/31/2025
100391	Washtucna	1,050	05/16/2024	08/31/2025
100392	Kahlotus	1,506	05/24/2023	08/31/2025
100393	Mossyrock	684	09/14/2023	08/31/2025
100394	Cusick	2,050	09/21/2023	08/31/2025
100395	Plain	1,868	10/13/2023	08/31/2025
100396	North Bend	1,556	09/14/2023	08/31/2025
100397	Northport	1,529	09/21/2023	08/31/2025
100398	Davenport	2,490	10/04/2023	08/31/2025
100399	Oroville	997	10/13/2023	08/31/2025
100400	Okanogan	1,046	10/13/2023	08/31/2025
100401	Chimacum	120	10/26/2023	08/31/2025
100402	Paterson	387	11/02/2023	08/31/2025
100418	Clyde	1,386	05/16/2024	08/31/2025
100419	Sunnyside	683	06/05/2024	08/31/2025
100421	Shelton	194	06/27/2024	08/31/2025
300205	Hartline	1,840	10/04/2023	08/31/2025
300208	Dayton	2,295	05/17/2023	08/31/2025
300219	Addy	1,707	09/21/2023	08/31/2025
310022	Montesano	24	06/27/2024	08/31/2025
310023	Roosevelt	400	10/27/2022	08/31/2025
310031	Toppenish	783	06/05/2024	08/31/2025
330035	Chelan	1,255	10/13/2023	08/31/2025
330036	Thorp	1,770	08/16/2023	08/31/2025
330040	Ephrata	1,224	07/29/2021	08/31/2025
330115	Sunrise	875	07/14/2021	08/31/2025

References

Cini, A.; and Marisca, I. 2022. Torch Spatiotemporal.